

# A Practical Sanity Check of GiBleed

*Recovering an undocumented disease module from a synthetic-data generator's output — using only SQL and clinical reasoning.*

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## Summary

Using seven SQL queries and clinical reasoning, I audited GiBleed — the Synthea-generated [2] Eunomia dataset [1] on which most people first learn the OMOP Common Data Model — and recovered an **undocumented generator module from its output alone**, without reading Synthea's source or documentation. Eight conditions occur exactly once per person and never again; seven of them, read together, are the **differential diagnosis of gastrointestinal (GI) bleeding**, and the eighth is osteoarthritis, present in 100% of the population. All eight are confined to a ~15-year midlife window and stop by age 47, in a database that records other conditions to age 109 and in which **no one ever dies**.

None of this violates a range check. Every value is well-formed, in-range, type-correct, and conformant — the dataset passes the Kahn et al. conformance/completeness/plausibility framework [6] and the OHDSI Data Quality Dashboard. The artifacts are invisible to structural tooling and visible only to a combined clinical-biomathematical reading.

**The thesis in one line.** The mathematics tells me what shape a disease should leave in data; the medicine tells me when the shape is wrong.

This is a self-directed learning artifact, not a publication. Its purpose is to demonstrate, on the ecosystem's own canonical teaching dataset, the multidisciplinary sanity check that structure-only tooling cannot perform — and that a federated research network needs performed **at each source**.

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## 1 • Why I did this

I wanted to be genuinely comfortable inside the OMOP/OHDSI ecosystem — not merely familiar with it. The gap was practical rather than conceptual: I knew the epidemiology and the modelling, and I needed fluency with the stack. So I did what I would do with any unfamiliar instrument — I pointed it at something and looked, by performing an actual sanity check to gain hands-on experience.

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## 2 • The observation that started it

Within an hour of loading the Eunomia GiBleed dataset [1] — synthetic, realistic, and the default sandbox where most people first meet OMOP — one thing stopped me:

The dataset has 2,694 patients. It also has exactly 2,694 cases of osteoarthritis.

Osteoarthritis is a chronic, degenerative disorder of articular cartilage driven by

mechanical wear over time; its principal risk factors are age, obesity, female sex, and prior joint trauma [28]. At the population level, symptomatic osteoarthritis most frequently occurs after age 40, and prevalence rises steeply with age [8]. A 100% prevalence figure is therefore not a data point but a claim about a population — that *everyone here is old enough to have worn out a joint*.

That produced a prediction, before I ran a single further query:

|

If every person has a degenerative disease, this population should contain almost no paediatric patients.

The prediction held, and the mechanism was stranger than expected. Every osteoarthritis record in the database is written at age  $\geq 31$  (Table 2; Figure 3). Since all 2,694 people have one, all 2,694 people reached 31. There are no children in GiBleed — only childhood *records* belonging to adults who later grew up. One anomaly, one hypothesis, one confirmation. That was the whole beginning, and it told me a multidomain sanity check of this teaching dataset would let me learn the CDM by doing while producing something useful.

### 3 • The lens I brought

My assumption at the outset was that GiBleed would be structurally and numerically flawless. It is. But structural validity is one question; a representative dataset must also satisfy properties that automated conformance checks never test — chiefly **representativeness** of the source population, **absence of systematic bias**, and **independence** of observations [27]. Establishing those requires three overlapping disciplines, with *fitness for purpose* at the centre.

Table 1. The three-domain sanity-check framework.

| Domain           | The question it asks                     | GiBleed               |
|------------------|------------------------------------------|-----------------------|
| Computer science | Is the value well-formed?                | ✓ Passes (conformant) |
| Medicine         | Is this what the disease naturally does? | ✗ (this audit)        |
| Biomathematics   | Is this a shape a disease can leave?     | ✗ (this audit)        |

Automated data-quality tooling clears only the first circle. Every finding below sits in the **Medicine**  $\cap$  **Biomathematics** overlap, where no range check has ever looked. The centre — *fitness for purpose* — requires all three, and the OMOP CDM has no field for it. That intersection is what I bring to a research team, and it is the argument of this repository.

### 4 • The two questions

- **From medicine:** does this dataset behave like the disease process — in real-world prevalence, risk stratification, natural history, pathophysiology, prescribing, complications, and differential diagnosis?

- **From biomathematics:** treating each condition as a compartment in a deterministic system, what is the *topological shape* of the disease states and the *flow dynamics* between them? A condition's transition structure leaves a signature in the record counts. The instrument for this second question is a column most people treat as bookkeeping: **records per person**.

Everything below comes from seven SQL queries against the 2,694-person Eunomia database [1] (`sql/audit-queries.sql`). No machine learning, no privileged access, roughly ninety seconds.

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## 5 • Findings

### 5A • The medical lens

*Most consequential first. Each finding carries the downstream analytic damage it invites by departing from the population it purports to represent.*

#### M1 • There is no confounding by indication

355 of 479 GI-bleed patients are on celecoxib (74.1%), and in every case the drug precedes the bleed — 355 times out of 355, with no reversals and no same-day ties (Figure 5).

In the real world, patients prescribed celecoxib are precisely those at elevated GI risk: because a COX-2-selective agent spares the COX-1-dependent gastric mucosa, it is preferentially channelled to patients with prior gastropathy, gastroprotective-agent use, and advancing age [16]. Coxib users accordingly carry a greater underlying disease burden than users of nonselective NSAIDs [16]. This is the definition of **confounding by indication** [15] — the reason NSAID→GI-bleed is a hard causal question rather than a division problem.

**Downstream damage — the most serious finding in this audit.** Every standard confounding-control method runs cleanly on GiBleed: propensity scores [18], inverse probability of treatment weighting (IPTW) [19], new-user designs [20], negative controls [21], and self-controlled designs [22] all execute, converge, and return plausible numbers. They have to. Each of these methods exists to *subtract a bias*. GiBleed contains none, so each one subtracts zero and returns the number it was handed. **A method validated here has not been validated** — a correct implementation and a broken one produce the same result, because there is nothing to detect. Three consequences follow:

- **RR = 1.32 is not a measurement.** With no confounding and no noise, the risk ratio (OR = 1.40) reflects the generator's parameter settings — not biology, not bias, not celecoxib.
- **Benchmarking here is uninformative.** A new causal-inference method tested against GiBleed will appear to work; that conclusion is unsupported, because it shows the code executes, not that the method removes bias.
- **The diagnostics are untested too.** There are no positivity violations, no extreme weights, and no overlap failures — the pathologies these checks exist to catch never occur.

A dataset built to teach a method for handling confounding contains no confounding. It teaches the mechanics perfectly and the reason for the mechanics not at all.

**M2 • 100% osteoarthritis prevalence**

Every person carries osteoarthritis, with exactly one record each (2,694 of 2,694). Real-world symptomatic osteoarthritis prevalence is roughly an order of magnitude lower — on the order of ~13% among adults aged 55 and over [8].

**Downstream damage.** No unexposed comparator exists — there is no osteoarthritis-free group to compare against. With zero variance, osteoarthritis has zero covariance with every other variable; it cannot confound, mediate, stratify, or be adjusted for. **Formally, it is not a variable but a constant** — indeed, the sample space itself. Any model entering it as a covariate is rank-deficient in that dimension (the column is identical to the intercept), so its coefficient is not separable from the baseline. Every comorbidity overlap involving osteoarthritis is therefore forced: for example, Otitis media  $\cap$  Osteoarthritis = 2,025, which is *all* 2,025 otitis patients. The co-occurrence carries no information; it is arithmetically guaranteed and serves only to confirm the 100% figure and the integrity of the person-level joins.

**M3 • The GI-bleed differential is present, complete, and one-shot**

Eight conditions occur exactly once per person and never again. Read as a clinician, seven of them are the differential diagnosis of GI bleeding.

**Table 2. The eight one-shot conditions of the GI-bleed module.**

| GI section | Condition                   | People (n) | Prevalence | Age window | Role       |
|------------|-----------------------------|------------|------------|------------|------------|
| Upper GI   | Peptic ulcer                | 802        | 29.8%      | 24–34      | Cause      |
| Upper GI   | Esophagitis                 | 409        | 15.2%      | 31–44      | Cause      |
| Upper GI   | Angiodysplasia of stomach   | 388        | 14.4%      | 30–44      | Cause      |
| Lower GI   | Diverticular disease        | 405        | 15.0%      | 32–47      | Cause      |
| Lower GI   | Polyp of colon              | 380        | 14.1%      | 31–46      | Cause      |
| Lower GI   | Ulcerative colitis          | 413        | 15.3%      | 31–45      | Cause      |
| Outcome    | Gastrointestinal hemorrhage | 479        | 17.8%      | 32–47      | Outcome    |
| Indication | Osteoarthritis              | 2,694      | 100%       | 31–47      | Indication |

This is the list one generates at the bedside — instantiated as a family of one-shot events.

**Downstream damage.** Nobody bleeds twice. Rebleeding is *the* clinical problem in GI hemorrhage, and it does not exist here, so time-to-recurrence, rebleed risk, and secondary prevention are unstudyable. Each cause appears once, in isolation, with no severity gradient

and no progression, so etiology-specific risk cannot be modelled: ulcerative colitis with no flares, diverticular disease with no diverticulitis — despite ~20% of incident diverticulitis recurring in reality [10] — and peptic ulcer with no *H. pylori* eradication and no relapse.

**A note on two columns that are easily conflated.** `(recs_per_person = 1.00)` is not 100% prevalence. Ulcerative colitis affects 15.3% of people, and all 413 of them got it exactly once. Only osteoarthritis is at 100% prevalence.

#### M4 · The age windows are clinically improbable

Every module member is diagnosed inside a ~15-year midlife window and never after 47 — in a database that records other conditions out to 109.

- **Angiodysplasia of the stomach** is the most common vascular abnormality of the GI tract and the second leading cause (after diverticulosis) of lower-GI bleeding in patients over 60, typically in the context of aortic stenosis and acquired von Willebrand syndrome (Heyde syndrome) [11]. Here it appears in 388 people aged 30–44, and never again.
- **Diverticular disease** rises with age from roughly 5% at 40 to about 65% by 85 [10]. Here it stops at 47.
- **Polyp of colon** — colorectal screening now begins at 45 precisely because incidence climbs through and beyond that age [12]. Here polyps are finished by 46.
- **Osteoarthritis** rises steeply after 55 and does not switch off [8]. Here nothing new is recorded after 47.

This is not censoring, and the proof is in the same table: viral sinusitis is recorded from age 0 to 109 (Figure 3). The recording apparatus demonstrably works across the whole lifespan, so the module's silence after 47 is a true absence.

**Downstream damage.** Age cannot confound within the module, so age-adjusted models are unidentifiable; age-stratified analysis is meaningless because there are no strata above 47; incidence-by-age curves are step functions with an off switch, which no real degenerative or vascular process produces. Any study relying on age as an effect modifier will find no effect — and the null will be an artifact.

**A precision point I want to be exact about.** The pathology is *not* an inverted age gradient. Prevalence rises from 0% to  $k\%$  across the window and then plateaus — monotone increasing, exactly as a chronic, irreversible condition should behave. What is wrong is that **incidence** has an off switch and the window is shifted young and truncated at 47; **prevalence** does not fall, because no one recovers and no one dies.

#### M5 · No deaths are recorded

There are zero deaths, yet 98 people are over 100 (3.6%, against a real-world figure of ~0.03% [13]), and the population reaches 110. A population spanning every age to 110 with no death record is not plausible.

**Downstream damage.** Competing risk does not exist — in real GI-bleed research, death

competes with rebleeding, so cumulative incidence and naive risk diverge; here they coincide [24]. Survival analysis is vacuous because there are no events. Immortal time bias is undetectable because everyone genuinely is immortal [25]. No mortality-adjusted or death-censored design can be tested against this data.

**M6 • Prescribing does not resemble prescribing**

The dataset shows 68% celecoxib exposure and 98.7% tetanus/diphtheria toxoid coverage. Celecoxib is a second-line, cardiovascular-risk-carrying, selectively channelled COX-2 inhibitor [16]; at the population level, coxibs are a small, selected share of NSAID prescribing [17]. A 68% population exposure is therefore implausible by roughly an order of magnitude. Likewise, only about 59% of U.S. adults report a tetanus-toxoid-containing vaccination in the past decade [14], against 98.7% here.

**Downstream damage.** No one was switched to a COX-2-selective agent *because* they were high-risk, so channelling — the real-world driver of confounding by indication — is absent. One exposure record per person compresses years of chronic NSAID therapy, the mechanism by which GI risk accumulates, into a single event. And near-universal adherence makes adherence research and healthy-adherer bias unstudyable.

**M7 • There is no comorbidity structure**

Every pairwise overlap equals the smaller group entirely — a consequence of osteoarthritis behaving as a constant rather than a variable (M2).

**Downstream damage.** Charlson and Elixhauser comorbidity indices are effectively constants. Multimorbidity research is impossible, and covariate adjustment is vacuous because there is nothing to adjust for. In reality, a 70-year-old with osteoarthritis typically also carries hypertension, diabetes, or chronic kidney disease; here the conditions do not interlock.

**5B • The biomathematical lens**

*Each condition is treated as a compartment in a deterministic system, which implies transition dynamics between states.*

**B1 • Eight absorbing states that should not exist**

`recs_per_person` is not bookkeeping; it is a measurement of the disease's state graph. A value of 1.00 signifies an **absorbing state** — entered once, never re-entered, never left. For a chronic, recurrent, or progressive condition this is the wrong topological shape: the transition matrix has a single entry and no self-loop. The evidence is the contrast — most conditions recur, tracing the transitions a real disease process leaves behind.

**Table 3. Records per person across representative conditions.**

| Condition                       | Records per person |
|---------------------------------|--------------------|
| Viral sinusitis                 | 6.43               |
| Acute viral pharyngitis         | 3.92               |
| Acute bronchitis                | 3.22               |
| Otitis media                    | 1.78               |
| Sprain of ankle                 | 1.41               |
| Chronic sinusitis               | 1.02               |
| <b>The eight module members</b> | <b>1.00</b>        |

Twenty-three conditions behave like life; eight are pinned dead flat at 1.00 (Figure 1). One number alone proves nothing — the neighbours make it legible. Viral sinusitis at 6.43 is the positive control: the apparatus can represent recurrence, so the flatness of the eight is a design decision, not an instrument limit.

**Downstream limitations.** Multi-state and Markov models are unidentifiable, because no transition rates can be estimated when no transitions occur [23]; there are no rate parameters, sojourn times, or hazards of re-entry. Hierarchical and mixed models have no within-person variance — with one observation per person, the random-effect variance component cannot be separated from the residual, so random effects are unidentifiable. And because the join cardinality is 1:1, fan-out — the one-to-many record explosion every EHR analyst spends a career defending against — simply does not occur. Its absence is itself the artifact.

## B2 • The death compartment was never built

Death is the canonical absorbing state of any human-population model — every  $S \rightarrow I \rightarrow R$  system, every Markov chain over a lifespan, every survival model has one. GiBleed's is missing entirely while the population ages to 110.

**Downstream limitations.** This is an open system with no exit; population is not conserved, so no balance- or carrying-capacity model can be fitted, and any compartmental model with a mortality sink is unidentifiable [24]. The absence is invisible to a conformance check, because an empty table is a valid table.

## B3 • The hazard function has an off switch

For osteoarthritis, incidence is zero before ~31, positive inside the window, and zero after 47 — in a system that demonstrably records events to 109.

**Downstream limitations.** No parametric survival family can fit this. The Weibull, Gompertz, and exponential families all assume a hazard that is monotone or at least continuous [26]; a hazard that terminates is a rectangular pulse. Real degenerative and

vascular hazards accumulate with age, whereas this one is switched off by a gate. As noted in M4, the gradient is *not* inverted — prevalence rises and then plateaus, monotone and correct; the defect is that incidence has an off switch and the window is shifted young and truncated. A fitted survival model will nonetheless return parameters with standard errors — the best possible summary of a shape the family cannot represent — which is precisely the kind of silent, plausible-looking failure this audit exists to surface.

**B4 • Determinism where stochasticity belongs**

Exposure precedes outcome 355 times out of 355. The `(>=)` in the ordering test folds same-day cases into `(bleed_first)`, so zero means zero reversals *and* zero same-day ties. Real data carries noise, misclassification, backfilled coding, reverse-time artifacts, and same-day collisions; real ordering runs ~95-98% with a ragged tail. A perfect, tail-free result reveals an arrow that was drawn, not observed.

**Downstream limitations.** There is no measurement error to model and no misclassification to correct. Window-based cohort definitions — index dates, lookback periods, tie-breaking rules — are never stressed, and any method robust to temporal noise is untested here.

**B5 • Two independent signatures, one boundary, no overlap**

`(recs_per_person)` and the age envelope were measured separately, yet they select the same eight conditions.

**Table 4. Two independent signatures selecting the same eight conditions.**

| Signature              | The eight module members  | The other twenty-three |
|------------------------|---------------------------|------------------------|
| Records per person     | 1.00                      | 1.02 – 6.43            |
| Age window (max – min) | ~15 years (none after 47) | up to 96–110           |

Nothing sits between them except one partially-scripted case — *E. coli* UTI (1.15 records/person, ages 16–66). Two unrelated rulers drawing the identical line do not describe a disease; they describe a code path (Figure 2).

**5C • Reverse-engineering Synthea's internal thresholding**

I never read Synthea's source code, module definitions, or documentation. Everything below was recovered from output alone — from 65,332 condition records and seven queries.

**Table 5. Generator internals recovered from output alone.**



| Recovered                                                                                         | Evidence                                                                  |
|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| A discrete GI-bleed module of 8 conditions exists                                                 | Two independent signatures select the same 8, with no overlap             |
| Its membership is the clinical differential of GI hemorrhage (3 upper, 3 lower, plus the outcome) | The identity of the eight                                                 |
| One-shot semantics — no recurrence loop                                                           | <code>recs_per_person</code> = 1.00, exactly, eight times                 |
| An onset age gate at ~30-32                                                                       | min ages: 30, 31, 31, 31, 31, 32, 32                                      |
| A termination gate at ~44-47                                                                      | max ages: 44, 44, 45, 46, 47, 47, 47                                      |
| Peptic ulcer runs a distinct, younger gate (24-34)                                                | ~7 years below the rest of the module                                     |
| Osteoarthritis is the eligibility scaffold, not a modelled disease                                | 100% prevalence + identical topology + identical window                   |
| No mortality module is wired into this scenario                                                   | <code>death</code> empty; ages to 110                                     |
| Module boundaries are hard, not probabilistic                                                     | <i>E. coli</i> UTI at 1.15 is the only intermediate case in 31 conditions |

I then checked the metadata. The `cdm_source` table independently confirms the generator: Synthea [2], OMOP CDM v5.3.1, ETL-Synthea [4], released 2019-05-25. The design-intent reconstruction — inferred from clinical reasoning alone, before reading the metadata — was correct:

```

100% osteoarthritis      -> the INDICATION (everyone NSAID-
eligible)
68% celecoxib + aspirin + ibuprofen -> the EXPOSURES
the GI differential, all one-shot    -> the ETIOLOGIES
gastrointestinal hemorrhage         -> the OUTCOME
the dataset is literally named GiBleed

```

The dataset is a sentence. Read the topology and you can read the design. (The Synthea authors have themselves published an erratum acknowledging unexplained age- and sex-related inconsistencies in generated data [3], which independently supports the general point that shape artifacts can survive documentation.)

## 5D • Why this matters in a federated ecosystem

This is where the argument stops being about GiBleed. OHDSI's method is federated [5]: an analytic package is written once and executed at many sites against local CDM instances. Patient-level data never moves; only aggregate results return. That architecture has a

specific blind spot. If a site's ETL introduces a topology artifact — collapsing recurrent encounters into one record, gating an age range, dropping a mortality feed — the network cannot see it. The site's data passes conformance, the results come back well-formed, and cross-site heterogeneity gets attributed to *population* differences when it is actually *ETL* differences. That is a confounder introduced by infrastructure, invisible to every tool currently pointed at the problem.

The implication: **data quality must be established at the source, not rescued during analysis.** A multidomain sanity check — structure *and* pathophysiology *and* topology — run at each node catches these artifacts before they enter the federation. Run afterwards, all one can do is discover that sites disagree and guess why. This is why the intersection in the Venn is a staffing argument, not a personal one. A CS-trained data-quality engineer will pass this dataset, correctly, by every standard in the toolkit; a clinician without modelling will see the ages are wrong but not that the shape is wrong; a modeller without medicine will see `(recs_per_person = 1.00)` with no prior that it should be 6. A team needs all three.

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## 6 • What GiBleed is, and is not

None of this is a bug. GiBleed was built to teach one thing — the NSAID→GI-bleed method — and every artifact here is scaffolding holding that lesson up. For its intended purpose it is excellent: small, fast, portable, conformant, and clean. This audit is not a criticism of Synthea, of Eunomia, or of the OHDSI community that maintains them.

The `(cdm_source)` table is exemplary provenance metadata — complete, honest, and well-linked; it independently confirmed my reconstruction, and it documents the data as "*synthetic, realistic (but not real)*." But it never says realistic *for what*. No field records that osteoarthritis was set to 100% as scaffolding, that `(death)` is empty, that GI hemorrhage is enriched, or that confounding was never generated. The CDM answers "*where did this come from?*"; it has no field that answers "*what can I use it for?*" (A minor related note: the bundled `(vocabulary_version)` is v5.0, dated 18-JAN-19 — the vocabulary everyone learning on Eunomia is taught against.)

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## 7 • Prior work

The broad claim — that synthetic health data lacks clinical realism — is not new, and I do not claim it. Chen et al. validated Synthea against real-world clinical quality measures in a 1.2-million-person Massachusetts cohort and found limited ability to model heterogeneous post-intervention outcomes [7]; Wagner and Blacketer (OHDSI Symposium, 2024) compared 1.16 million Synthea-generated California records to real-world data via Achilles and found limited modelling diversity, especially for chronic disease; and the conformance/completeness/plausibility framework GiBleed passes is itself well established [6].

What I believe is narrower and new is fourfold: (1) an audit of **GiBleed specifically** — the 2,694-person default teaching dataset, where most practitioners' mental model of "what OMOP data looks like" is formed, rather than the million-person populations prior work used; (2) **disease-state topology as a data-quality prior** — records-per-person as a

signature of transition structure, derived from compartmental-modelling formalism; (3) **module-boundary recovery from output alone** — an undocumented generator module identified via two independent clinical signatures, without source access; and (4) **design-intent reconstruction as a validation method** — recovered from clinical reasoning and then tested against the documented source. Because Synthea's design is documented, this last claim is falsifiable. It held.

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## 8 • Limitations

- **Real-world rates are order-of-magnitude clinical anchors, not the argument.** The point is scale, not decimals. Two figures are, however, exact: the centenarian share (~0.03% [13]) and adult tetanus-vaccination coverage (~59% [14]).
  - **n = 2,694.** These findings describe *this* dataset, not Synthea in general; prior work covers Synthea at scale [7].
  - **Osteoarthritis and angiodysplasia do occur in midlife** [9,11]. The impossibility is not their presence in the 30s and 40s but the combination of 100% prevalence, hard truncation at 47 in a population living to 110, and complete absence of recurrence.
  - **GI hemorrhage was identified by** `condition_concept_id = 192671` without descendant expansion via `concept_ancestor`.
  - **The module was identified from output only.** I have not read Synthea's source to confirm it corresponds to a discrete generator component. That is the falsifiable part, and I would welcome being checked.
- 

## 9 • A note on corrections

An earlier draft of this audit described the module's age gradient as *inverted*. That was wrong: prevalence is monotone-increasing but shifted young and truncated. I caught and corrected it after re-examining the incidence-versus-prevalence distinction (§M4, §B3). I record the correction here deliberately — applying the same standard to one's own work, in public, is the discipline the audit is meant to demonstrate.

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## Figures

Rendered from verified query output; see `figures/`.

- **Figure 1 —** `01-topology.png`. **Records per person, all 31 conditions.** Twenty-three conditions scatter between 1.02 and 6.43; eight sit pinned at the 1.00 floor. The contrast, not the value, is the evidence.
- **Figure 2 —** `08-module-boundary.png`. **Two independent signatures, one boundary, no overlap.** Each bubble is a condition (x = records/person, y = age envelope). Two unrelated rulers draw the same line.
- **Figure 3 —** `02-age-envelope.png`. **Age at diagnosis, all 31 conditions.** Twenty-three span a lifetime to 96–110; the eight occupy a ~15-year band and stop at 47. Viral sinusitis (0–109) is the positive control.

- **Figure 4** — [03-reality-gap.png](#). Observed versus real-world prevalence (log scale). GiBleed prevalences against order-of-magnitude clinical reference values; deaths omitted, as zero has no logarithm.
- **Figure 5** — [05-design-intent-flow.png](#). The dataset as a flow: indication → exposure → outcome. All 2,694 people enter as one mass because 100% carry the indication; note the 124 who bleed unexposed.
- **Figure 6** — [07-verdict-matrix.png](#). Fitness-for-purpose verdict matrix. Excellent, unusable, or dangerous, depending on the question — "dangerous" being the only silent failure.

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## Reproduce it

```
r

install.packages(c("Eunomia", "DBI", "RSQLite"))
library(Eunomia); library(DBI)
con <- dbConnect(RSQLite::SQLite(), getEunomiaConnectionDetails()$server())

# The query that found the module
dbGetQuery(con, "
SELECT c.concept_name,
       COUNT(*)                               AS n_records,
       COUNT(DISTINCT co.person_id)            AS n_people,
       ROUND(COUNT(*) * 1.0 / COUNT(DISTINCT co.person_id), 2) AS recs_per_person,
       MIN(CAST(strftime('%Y', co.condition_start_date, 'unixepoch') AS INTEGER)
       MAX(CAST(strftime('%Y', co.condition_start_date, 'unixepoch') AS INTEGER)
FROM condition_occurrence co
JOIN concept c ON co.condition_concept_id = c.concept_id
JOIN person p ON co.person_id = p.person_id
GROUP BY c.concept_name
HAVING COUNT(DISTINCT co.person_id) >= 300
ORDER BY recs_per_person DESC;
")
dbDisconnect(con)
```

*Portability note.* Eunomia's SQLite stores dates as Unix epoch integers rather than the CDM-specified `DATE` type — a small conformance deviation, found by checking a column before trusting arithmetic on it. Hence `strftime('%Y', col, 'unixepoch')`. On PostgreSQL use `EXTRACT(YEAR FROM col)`; on SQL Server, `YEAR(col)`.

Full notebook: [notebooks/gibleed-audit.ipynb](#) · Standalone SQL: [sql/audit-queries.sql](#)

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## References

1. DeFalco F, Schuemie M, Sena A, Adulyanukosol N, Liu S, Black A. **Eunomia: Standard Dataset Manager for OMOP CDM Sample Datasets**. R package v2.1.0. OHDSI; 2025.

<https://github.com/OHDSI/Eunomia>

2. Walonoski J, Kramer M, Nichols J, et al. **Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record.** *J Am Med Inform Assoc.* 2018;25(3):230–238. doi:10.1093/jamia/ocx079
3. **Erratum: Synthea** [age and sex inconsistencies acknowledged]. *J Am Med Inform Assoc.* 2018;25(7):921. doi:10.1093/jamia/ocx147
4. OHDSI. **ETL-Synthea: Synthea-to-OMOP-CDM extract-transform-load.** <https://github.com/OHDSI/ETL-Synthea>
5. Observational Health Data Sciences and Informatics. **The Book of OHDSI.** 2021. <https://ohdsi.github.io/TheBookOfOhdsi/>
6. Kahn MG, Callahan TJ, Barnard J, et al. **A harmonized data quality assessment terminology and framework for the secondary use of electronic health record data.** *EGEMS (Wash DC).* 2016;4(1):1244. doi:10.13063/2327-9214.1244
7. Chen J, Chun D, Patel M, Chiang E, James J. **The validity of synthetic clinical data: a validation study of a leading synthetic data generator (Synthea) using clinical quality measures.** *BMC Med Inform Decis Mak.* 2019;19(1):44. doi:10.1186/s12911-019-0793-0
8. GBD 2021 Osteoarthritis Collaborators. **Global, regional, and national burden of osteoarthritis, 1990–2020 and projections to 2050: a systematic analysis for the Global Burden of Disease Study 2021.** *Lancet Rheumatol.* 2023;5(9):e508–e522. doi:10.1016/S2665-9913(23)00163-7
9. He Y, Jiang W, Wang W. **Global burden of osteoarthritis in adults aged 30 to 44 years, 1990 to 2019: results from the Global Burden of Disease Study 2019.** *BMC Musculoskelet Disord.* 2024;25:317. doi:10.1186/s12891-024-07442-w
10. Strate LL, Morris AM. **Epidemiology, pathophysiology, and treatment of diverticulitis.** *Gastroenterology.* 2019;156(5):1282–1298.e1. doi:10.1053/j.gastro.2018.12.033
11. Hudzik B, Wilczek K, Gasior M. **Heyde syndrome: gastrointestinal bleeding and aortic stenosis.** *CMAJ.* 2016;188(2):135–138. doi:10.1503/cmaj.150194
12. US Preventive Services Task Force. **Screening for colorectal cancer: US Preventive Services Task Force recommendation statement.** *JAMA.* 2021;325(19):1965–1977. doi:10.1001/jama.2021.6238
13. Pew Research Center. **U.S. centenarian population is projected to quadruple over the next 30 years.** 2024. (*Analysis of U.S. Census Bureau data; centenarians ≈ 0.03% of the U.S. population.*) See also U.S. Census Bureau, *Centenarians: 2020, 2025.*
14. Centers for Disease Control and Prevention. **Vaccination coverage among adults in the United States, National Health Interview Survey, 2022.** AdultVaxView. (*59.2% of adults ≥19 reported a tetanus-toxoid-containing vaccination in the past 10 years.*)
15. Walker AM. **Confounding by indication.** *Epidemiology.* 1996;7(4):335–336. And Kyriacou DN, Lewis RJ. **Confounding by indication in clinical research.** *JAMA.*

16. **Prescription channeling of COX-2 inhibitors and traditional nonselective NSAIDs: a population-based case-control study.** *Arthritis Res Ther.* 2005. PMC1065326. And **Factors associated with celecoxib and rofecoxib utilization.** PMID:15755796. (*Coxib users carry greater underlying GI-risk and disease burden.*)
  17. **Prescriptions for chronic high-dose cyclooxygenase-2 inhibitors are often inappropriate and potentially dangerous.** *J Gen Intern Med.* 2005. PMID:16191131. (*Nonaspirin-NSAID and coxib population use anchors the order-of-magnitude comparison.*)
  18. Rosenbaum PR, Rubin DB. **The central role of the propensity score in observational studies for causal effects.** *Biometrika.* 1983;70(1):41–55. And Austin PC. **An introduction to propensity score methods for reducing the effects of confounding in observational studies.** *Multivariate Behav Res.* 2011;46(3):399–424. doi:10.1080/00273171.2011.568786
  19. Robins JM, Hernán MA, Brumback B. **Marginal structural models and causal inference in epidemiology.** *Epidemiology.* 2000;11(5):550–560. And Austin PC, Stuart EA. **Moving towards best practice when using inverse probability of treatment weighting (IPTW).** *Stat Med.* 2015;34(28):3661–3679. doi:10.1002/sim.6607
  20. Ray WA. **Evaluating medication effects outside of clinical trials: new-user designs.** *Am J Epidemiol.* 2003;158(9):915–920. doi:10.1093/aje/kwg231
  21. Lipsitch M, Tchetgen Tchetgen E, Cohen T. **Negative controls: a tool for detecting confounding and bias in observational studies.** *Epidemiology.* 2010;21(3):383–388. doi:10.1097/EDE.0b013e3181d61eeb
  22. Whitaker HJ, Farrington CP, Spiessens B, Musonda P. **Tutorial in biostatistics: the self-controlled case series method.** *Stat Med.* 2006;25(10):1768–1797. doi:10.1002/sim.2302
  23. Jackson CH. **Multi-state models for panel data: the msm package for R.** *J Stat Softw.* 2011;38(8):1–28. doi:10.18637/jss.v038.i08. And Andersen PK, Keiding N. **Multi-state models for event history analysis.** *Stat Methods Med Res.* 2002;11(2):91–115.
  24. Fine JP, Gray RJ. **A proportional hazards model for the subdistribution of a competing risk.** *J Am Stat Assoc.* 1999;94(446):496–509. And Austin PC, Lee DS, Fine JP. **Introduction to the analysis of survival data in the presence of competing risks.** *Circulation.* 2016;133(6):601–609. doi:10.1161/CIRCULATIONAHA.115.017719
  25. Suissa S. **Immortal time bias in pharmacoepidemiology.** *Am J Epidemiol.* 2008;167(4):492–499. doi:10.1093/aje/kwm324
  26. Collett D. **Modelling Survival Data in Medical Research.** 3rd ed. Boca Raton: CRC Press; 2015.
  27. Rothman KJ, Greenland S, Lash TL. **Modern Epidemiology.** 3rd ed. Philadelphia: Lippincott Williams & Wilkins; 2008.
  28. Kumar V, Abbas AK, Aster JC. **Robbins & Cotran Pathologic Basis of Disease.** 10th ed. Philadelphia: Elsevier; 2020.
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**The mathematics tells me what shape a disease should leave in data. The medicine tells me when the shape is wrong.**